

COGNITIVE COMPUTING

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- Foundations of Cognitive Computing -
- Design Principles of Cognitive System
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-



Introduction



- **Cognitive Computing represents self-learning systems that utilize machine learning models to mimic the way brain works.**
- **Eventually, this technology will facilitate the creation of automated IT models which are capable of solving problems without human assistance.**
- **The Cognitive computing is the imitation of the thought process of human beings using a sophisticated computerized model.**

- ▶ **Cognitive computing works on the principle of**
 - ❑ **Learn**
 - ❑ **Model**
 - ❑ **Generate Hypothesis**



Learn—A cognitive system learns. The system leverages data to make inferences about a domain, a topic, a person, or an issue based on training and observations from all varieties, volumes, and velocity of data.

Model—To learn, the system needs to create a model or representation of a domain (which includes internal and potentially external data) and assumptions that dictate what learning algorithms are used.

Understanding the context of how the data fits into the model is key to a cognitive system.

Generate hypotheses—A cognitive system assumes that there is not a single correct answer. The most appropriate answer is based on the data itself. Therefore, a cognitive system is probabilistic. A hypothesis is a candidate explanation for some of the data already understood.

A cognitive system uses the data to train, test, or score a hypothesis.

- ▶ **Cognitive computing allows a wide variety of problems previously incapable of being solved by fixed program instructions to be calculable and estimable.**
- ▶ **In order to achieve this new level of computing cognitive systems must be :**
 - ❑ **Adaptive**
 - ❑ **Interactive**
 - ❑ **Iterative**
 - ❑ **Stateful**
 - ❑ **Contextual**





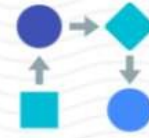
Adaptive



Interactive



Iterative



Stateful



Contextual

Flexible systems which learn as information changes, and as goals and requirements evolve.

System which interacts with other processes, devices, cloud services, as well as with people

Iterative process to solve problems which are ambiguous

Provides information that is suitable for the specific application at that point in time.

Understand, identify and extract contextual elements.

**The Purpose
of cognitive
computing is
to perform
tasks that only
humans used
to be able to.**



THE BASIS OF COGNITIVE COMPUTING

Big Data Analysis

Machine Learning



Cloud Computing

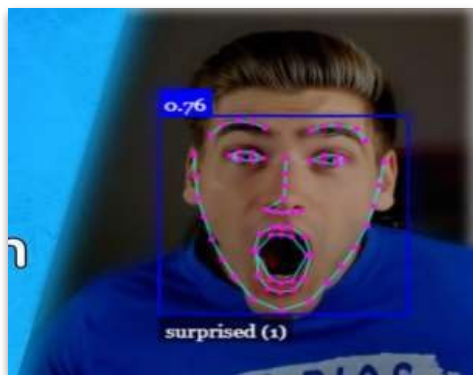
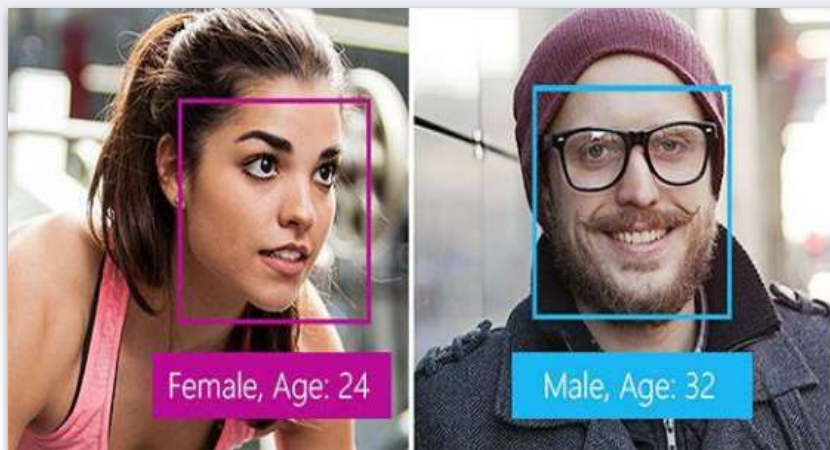
APPLICATIONS OF COGNITIVE COMPUTING



CHATBOTS



SENTIMENT ANALYSIS

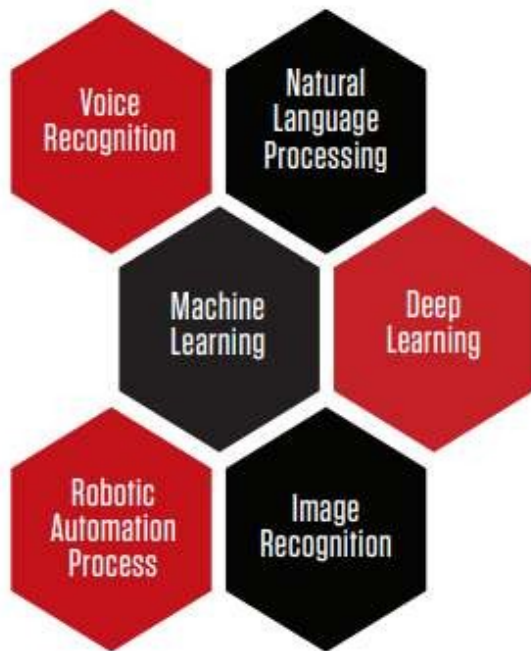


FACE DETECTION

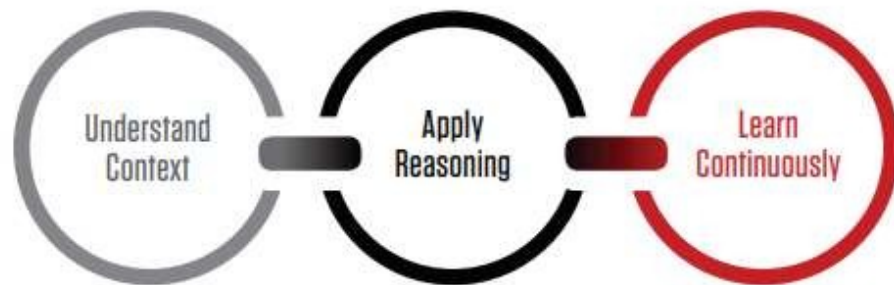


RISK ASSESSMENT

AI TOOLS



COGNITIVE ANALYTICS



---WORKING OF COGNITIVE
ANALYTICS---

COGNITIVE COMPUTING VS ARTIFICIAL INTELLIGENCE

CC	AI
• Cognitive computing learns & imitates the human thought process	• AI solve a problem through the use of best possible algorithm
• Cognitive computing doesn't throw humans out of the picture	• AI makes the final decisions at their own
• Augment human capabilities	• Automate processes
• Machine Learning,NLP,Neural network,cloud computing,big data analysis	• Machine Learning,NLP,Neural network,cloud computing,big data analysis
• Customer service,healthcare,industrial Sector	• Finance,security,retail,manufacturing,government

COGNITIVE COMPUTING LANDSCAPE

IBM WATSON



MICROSOFT COGNITIVE SERVICE



COGNITIVESCALE



SPARKCOGNITION



COGNITIVE COMPUTING FOR THE SOCIETY - USE CASES





1. ON EDUCATION

SYSTEM Cognitive computing will give rise to personal cognitive assistants for students, teachers and support staff.

- ▷ **Yearbooks, financial statement, student reports, and other regular documentation will be automatically handled by cognitive computing platforms.**
- ▷ **Personalised Notes and assessment materials**
- ▷ **Career guidance**
- ▷ **Act as a personal tutor, guiding students through their course work, explaining problematic sections.**
- ▷ **Personalised cognitive assistant for students who go to international universities to help them integrate with the place.**





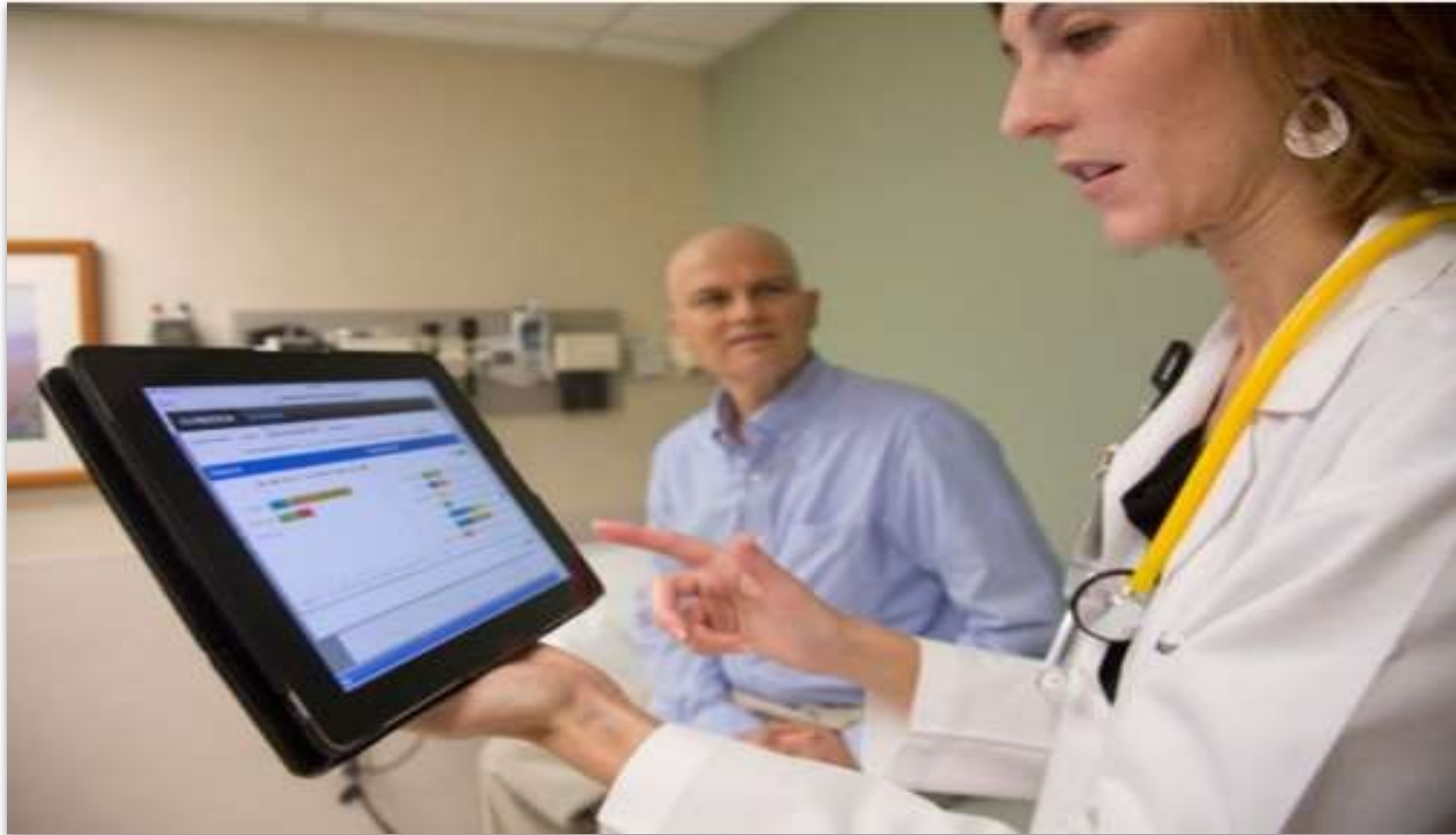
2. IN HEALTHCARE



- **Helps patients to get better treatment, have a quicker recovery, and lead improved lives, while doctors can diagnose better and save millions of lives.**

- **Faster Medical Research**
- **Improvement in daily processes**
- **Improved patient Interaction**
- **Diagnosis applications**





IBM Watson Cognitive computer assisting cancer diagnosis

ADVANTAGES OF COGNITIVE COMPUTING



- ▶ **Reduction in Human Error**
- ▶ **Prevention of wrong decision by machines.**
- ▶ **Available 24x7**
- ▶ **Helping in Repetitive Jobs**
- ▶ **Digital Assistance**
- ▶ **Faster Decisions**



DISADVANTAGES OF COGNITIVE COMPUTING



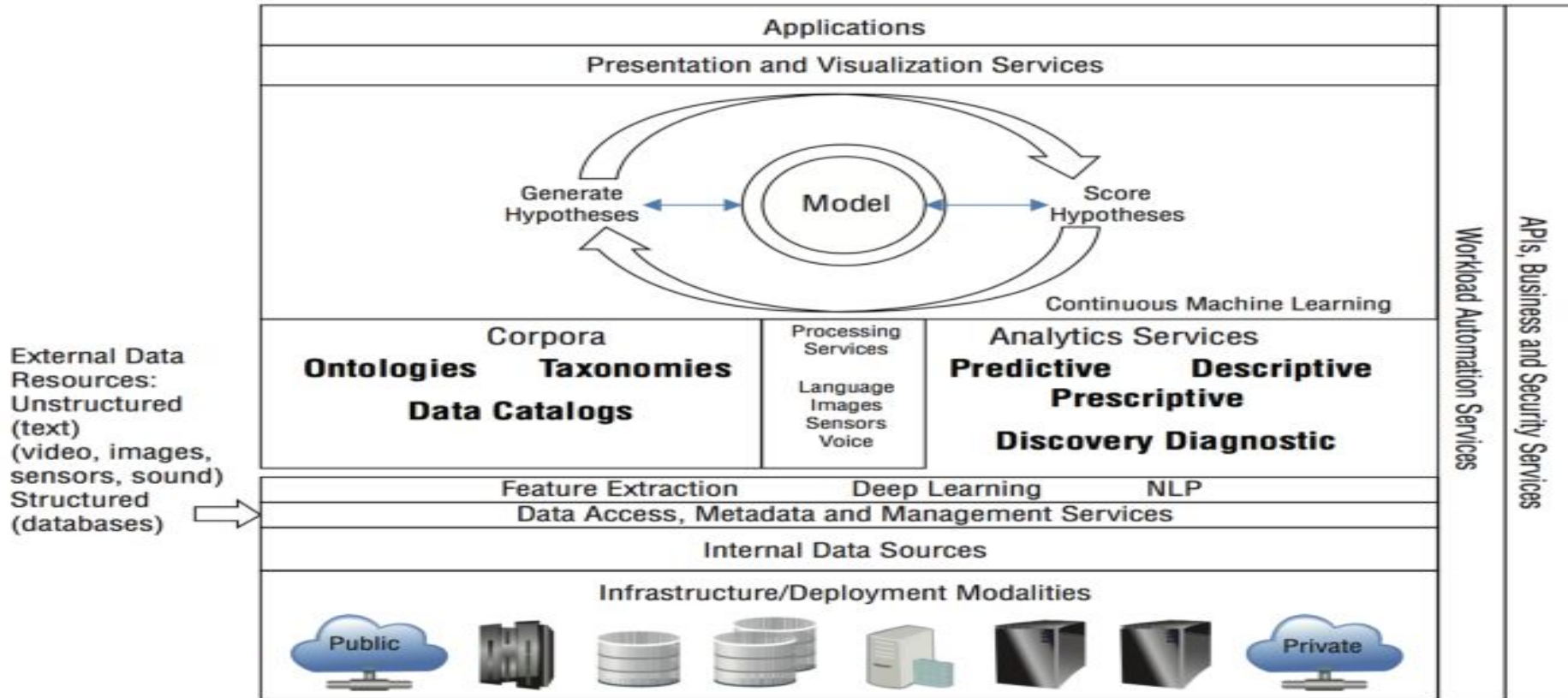
- ▶ **Security**
- ▶ **Change management is another challenge as this technology has power to learn like humans and behave like natural humans. Hence people are fearful as they feel machines would replace humans in future.**
- ▶ **Lengthy development cycles and Costly**
- ▶ **Making Humans Lazy**
- ▶ **No Emotions**
- ▶ **Unemployment**
- ▶ **Lacking Out of Box Thinking**

- ▶ **It will be in favor of all the organizations and humanity, at large, to start the transition process and adopt innovative technology for a bright and much more efficient future.**

▶



Elements of Cognitive Computing





.....Elements of CC

Infrastructure

Data Access , Metadata & Management Services

Corpus, Taxonomies & Data Catalogs

Data Analysis Services

Continuous ML

Hypothesis Generation and Evaluation

Tools and Learning Process

Presentation and Visualization

Cognitive Application



